Yue Su

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Summary

I build agent systems end to end: the knowledge graphs they retrieve from, the evaluations that measure them, and the tracing that shows what they cost. Shipped **Gene Quest** at Epsilla, contribute to **HarnessRouter**, published on agent memory bottlenecks and co-authored **ClawTrace**. I build the backend services around them. See more at yuesu0202.github.io.

Education

Carnegie Mellon University — *Aug 2024 – Dec 2025*
M.S. in Artificial Intelligence Engineering — *GPA: 3.93/4.00*

University of Illinois at Urbana-Champaign — *Aug 2020 – May 2024*
B.S. in Mathematics; B.S. in Statistics; Minor in Computer Science — *GPA: 3.88/4.00*

Experience

AI Engineer Intern — *Union City, NJ(Remote)*
Epsilla, Inc(YC S23) — *Feb 2026 – Oct 2026*

- Shipped **Gene Quest** as Next.js gene research agent over **Google Spanner** knowledge graph of **3.3M+** PubMed articles, **Neon Postgres** for product state and agents running on Claude Code harness. Exposed the graph to agents through **MCP** servers with 7 retrieval tools, and build app-side review loop to verify citations against the graph. Built gene monograph generation from ranked, evidence-graded claims, cached and shared across users.
- Co-authored and released **ClawTrace**, an open-source cost-aware tracing platform for LLM agents. Captured per-step cost and token attribution across LLM calls, tool calls, and sub-agents, compressed into compact TraceCards for skill distillation. Built an AI analyst that diagnoses expensive or failing runs and distills traces into reusable agent skills; showed cost-attributed prune rules reduce regressions from **14 to 5** versus an ablation without them.
- Contributed **HarnessRouter**, open-source router for 30 coding-agent harnesses. Moved DeepSeek Harness onto the shared relay and fixed self-hosted model failures by stripping oversized `max_tokens` and surfacing providers' refusal reasons.

Software Engineer Intern — *Shenzhen, China*
Meituan — *Jun 2024 – Aug 2024*

- Built microservice for batch PDF processing handling ingestion, processing, and delivery through an API and parallelizing work. Used **PyMuPDF** to parse text, images, layouts and CV model **Detectron2** detects document structure, improving extraction accuracy **22%**. Combined the extracted text with structure into a machine-readable document representation of typed blocks with coordinates and text to improve downstream search from structure JSON and highlighted document views for review. Containerized the services with Docker and deployed on Kubernetes.

AI Specialist (Part-time) — *Remote*
Handshake AI — *Jul 2026 – Nov 2026*

- Design and run evaluations of frontier coding agents on long-horizon tasks, measuring behavior including approval-request frequency, red-teamed agent policy boundaries on a simulated social platform, and documenting reproducible failure cases.

Research

Research Assistant — *Pittsburgh, PA*
Mechanical and AI lab, Carnegie Mellon University — *Nov 2024 – Dec 2025*

- **Distributed Fault-Tolerant System**: Designed distributed fault-tolerant system with active/passive replication, Redis deduplication, and heartbeat-driven failover, detecting failures within 1s, and auto-recovering within 3s without message loss.
- **Meta-Learning vs Finetune on Protein Mutation Sequence**: Developed a PyTorch MAML framework for multi-task protein mutation sequence prediction with 16-shot adaptation, modular training across multiple protein datasets, and input sequence encoding, achieving 46% lower MSE, 10% higher accuracy, and 73% lower training time than finetuning baselines.
- **Generative AI-Powered Document Automation**: Built end-to-end platform that converts PDFs into structured PowerPoint slides and narrated videos, with Python REST APIs React frontend for upload/status tracking, and AWS Lambda, EC2, S3, and SQS orchestration to reduce processing time to under 5 minutes per document and compute cost by 35%.

Publications

ClawTrace: Cost-Aware Tracing for LLM Agent Skill Distillation *ACM CAIS 2026 Workshop on Agent Skills (Oral)*
Diagnosing Retrieval vs. Utilization Bottlenecks in LLM Agent Memory *ICLR 2026 Workshop on MemAgents*

Skills

Languages: Python, C++, Java, SQL, JavaScript/TypeScript, Bash, Rust
Post-training: PyTorch, SFT, LoRA, RL, RLHF, Hugging Face
AI/ML: Transformers, **RAG**, **LangGraph**, LangChain
Backend & Systems: FastAPI, Next.js, REST APIs, LLM APIs (OpenAI, Claude, Gemini), Distributed Systems, React, Spark
Cloud & Infrastructure: AWS, **GCP**, **Docker**, **Kubernetes**, CI/CD, **Git**, **Linux**